LaharZ Example

A) <u>Model setup</u>

1. Login to VICTOR through `hub.victorproject.org` in your browser
2. Choose a small machine
3. Once logged in, open a Terminal window, type `victor setup` and choose `LaharZ`
4. Copy the DEM of Mt Rainier named `Rainier.tif` from `~/shared/DEMs/NorthAmerica/` into the new folder `LaharZ/example`
5. Launch the Desktop using the 

icon

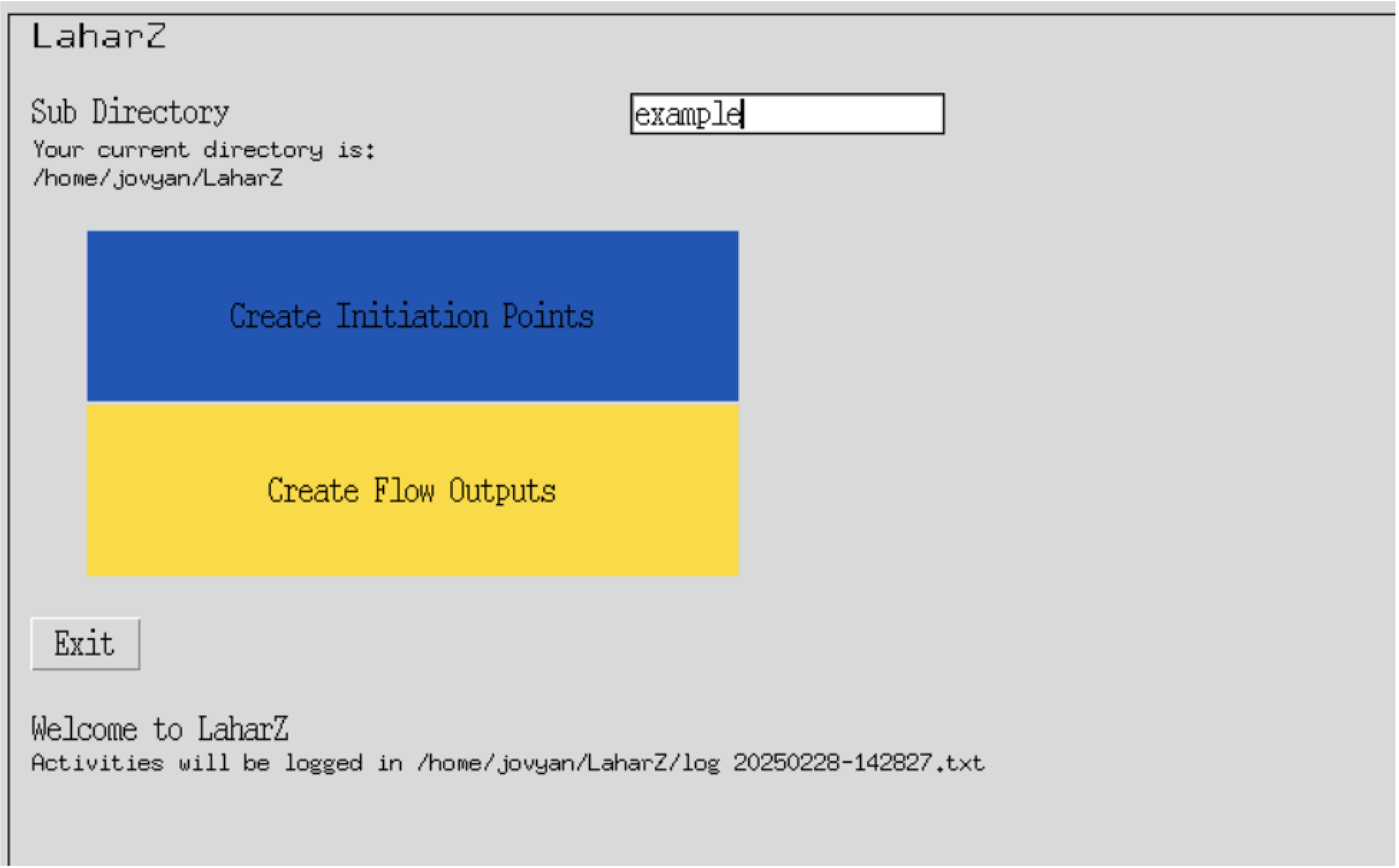
6. Open a Terminal window on the desktop by clicking the 

icon

7. Change directory into the folder `LaharZ` by typing `cd ~/LaharZ` in the terminal window and pressing enter.

8. Run the program by typing *`python laharz.py`* and pressing enter. A new window will appear.

In the GUI, type “`example`” for the folder name, and click `Create Initiation Points`

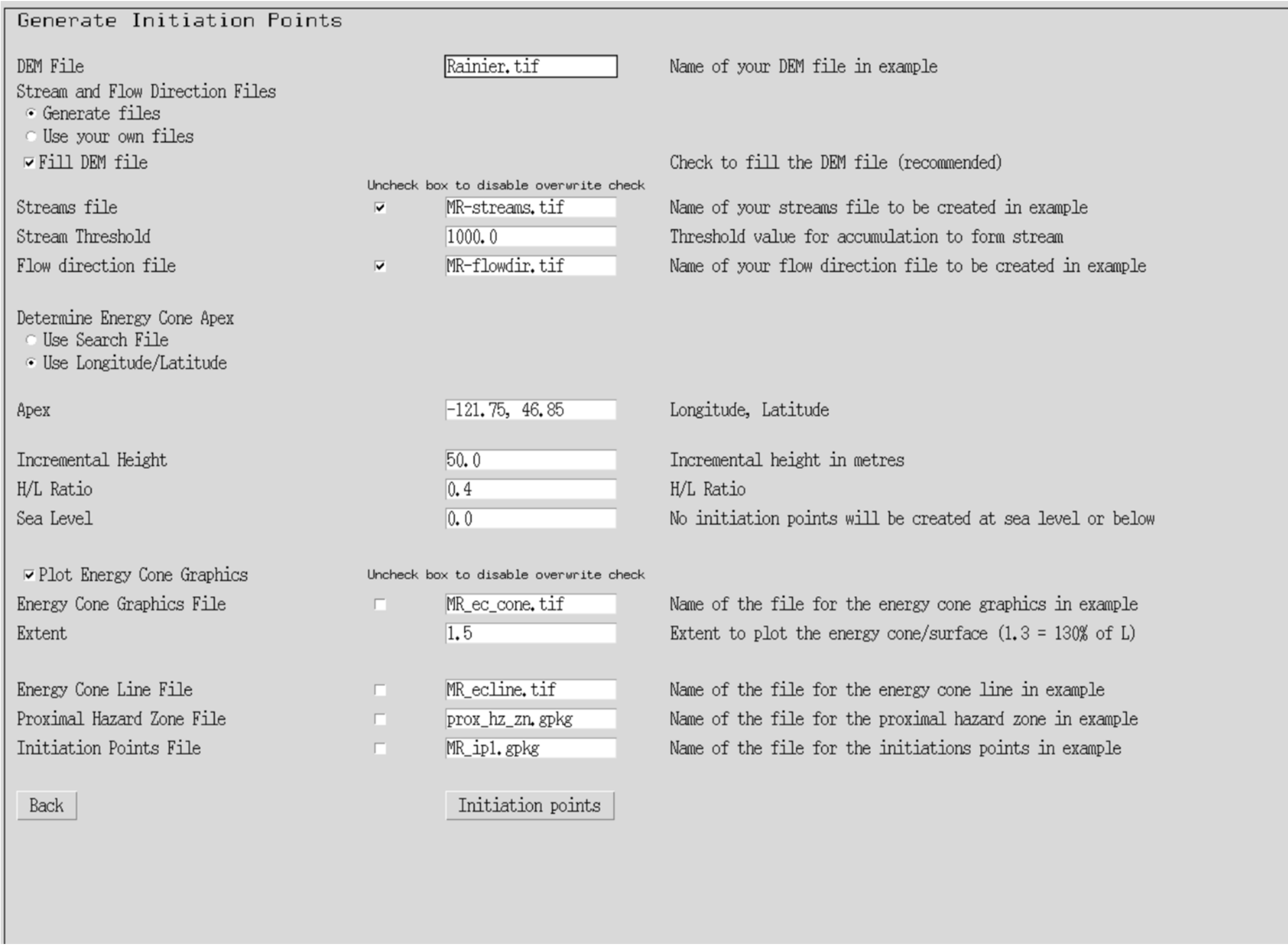


A) Creating initiation points:

1. Enter the input parameters for the initiation points

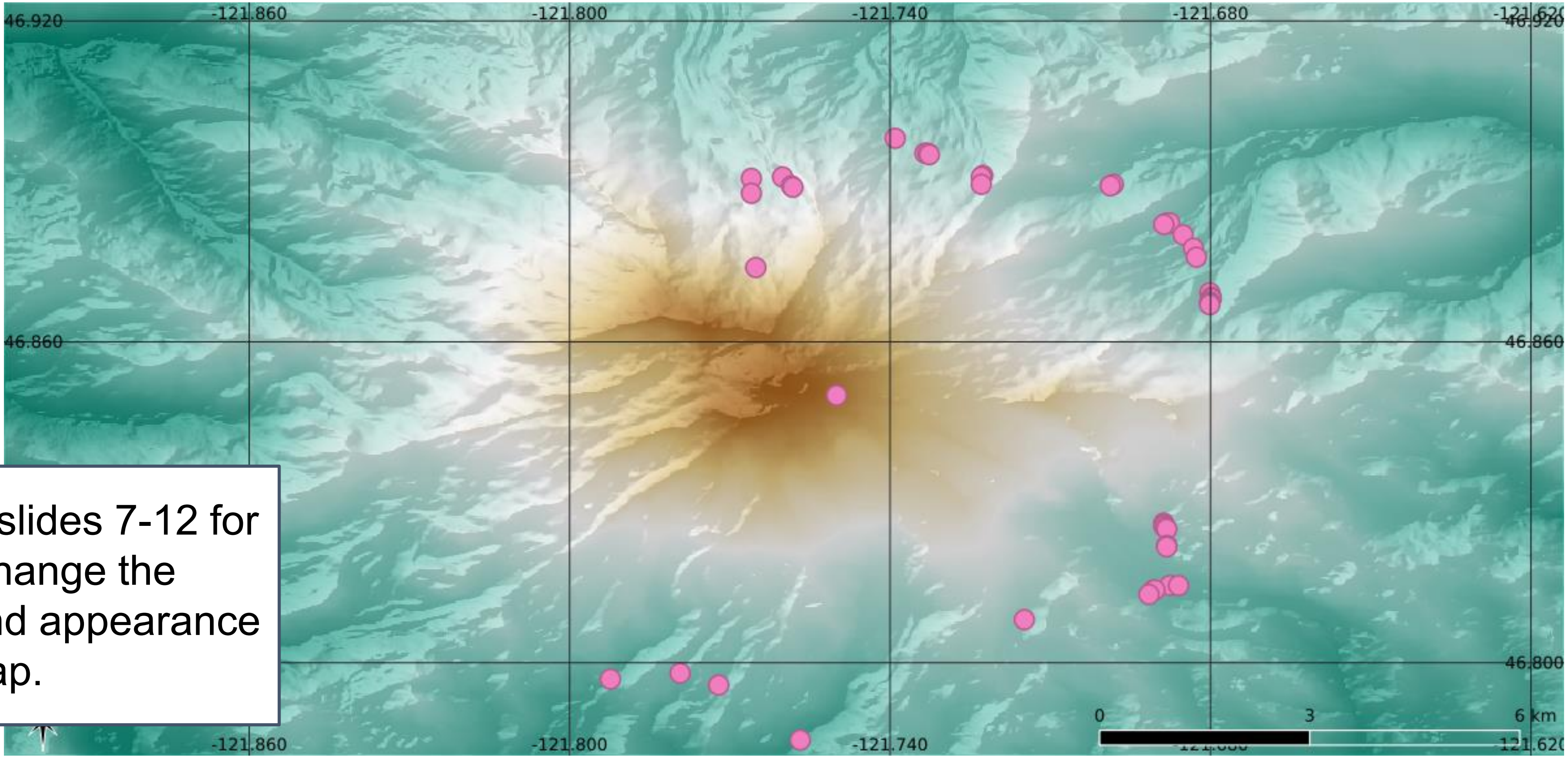
1. Click the “Initiation points” button

1. If asked, click “overwrite” button



Visualization of Initiation Points

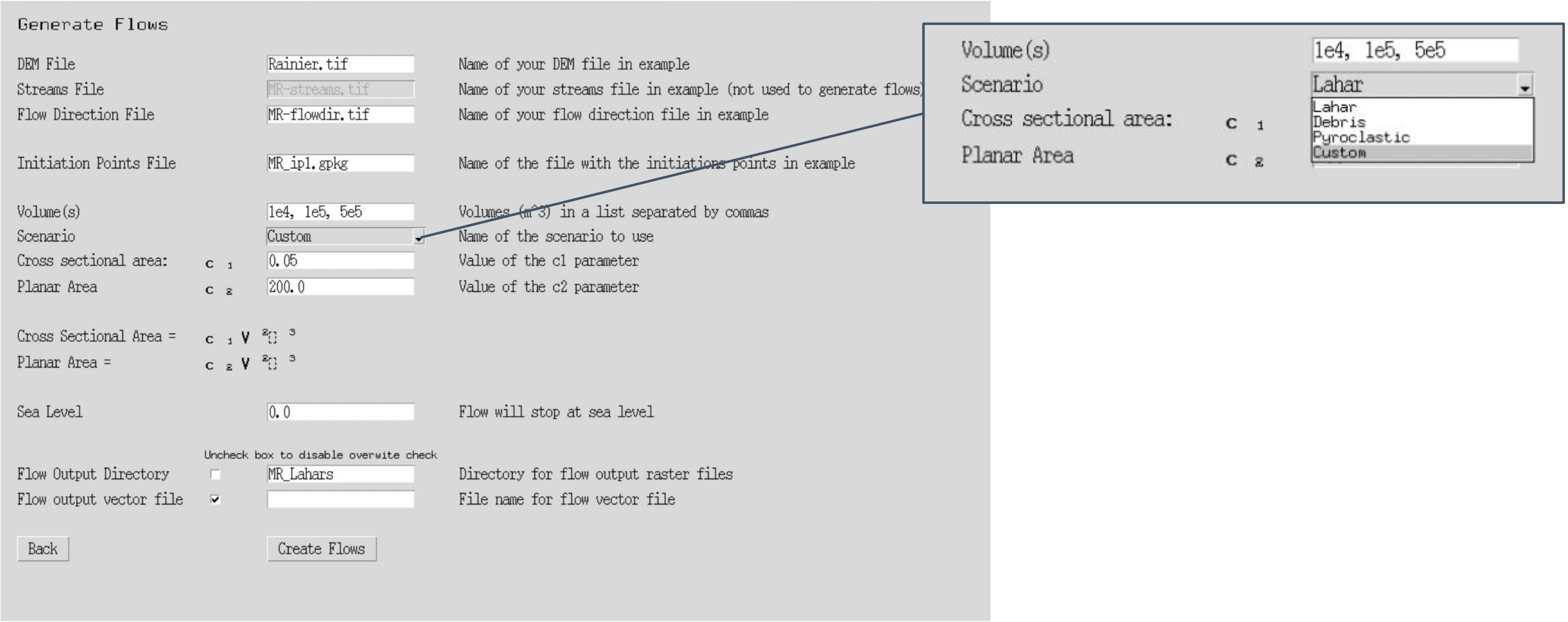
1. On the Desktop, open QGIS
2. Load the Mt Rainier DEM using "`Layer->Add Layer->Add Raster Layer`"
3. Add the initiation points just created using "`Layer->Add Layer->Add Vector Layer`" and loading the file `MR_ip1.gpkg`



❗ See slides 7-12 for how to change the colors and appearance of the map.

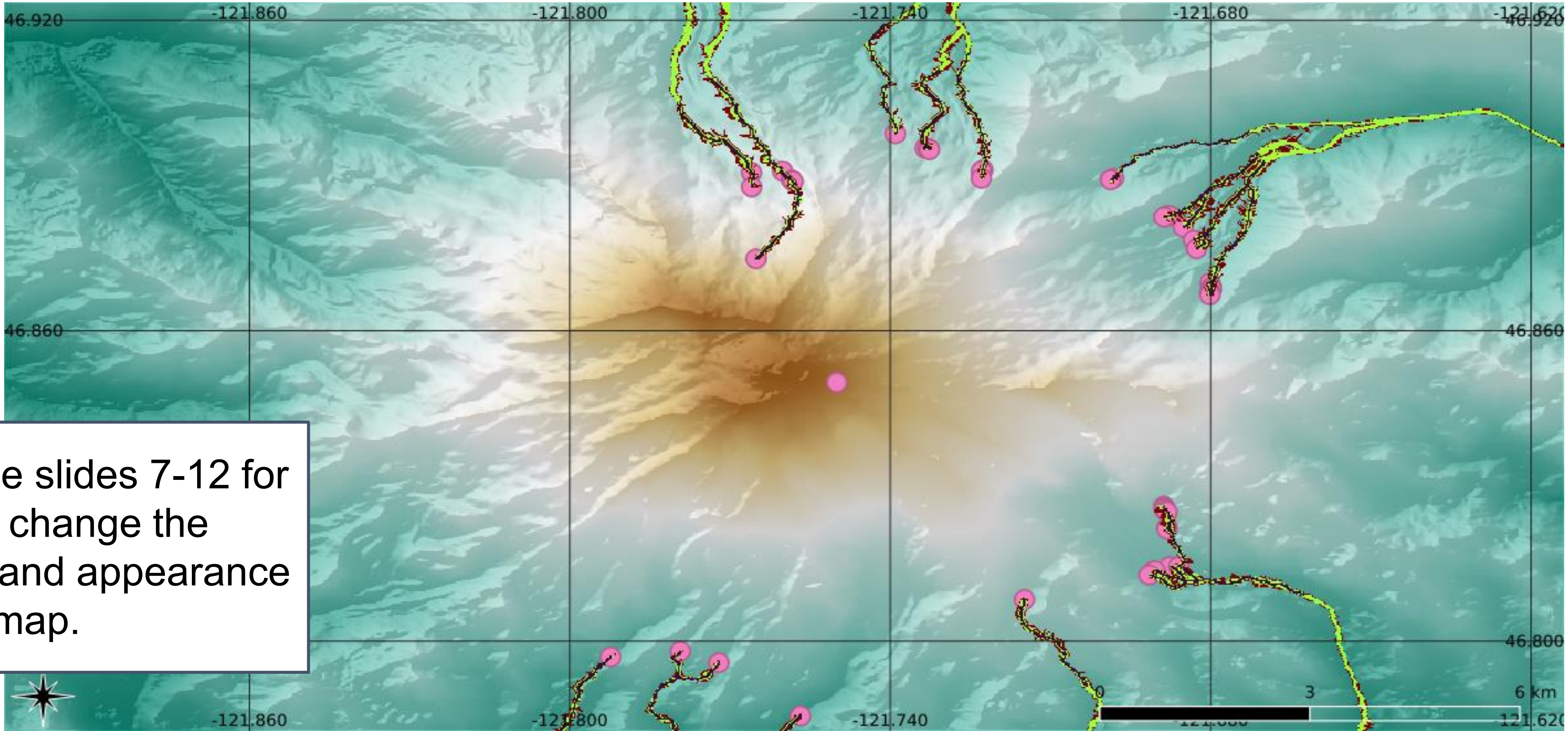
B) Run the flow model

1. In the GUI, click and run Create flow outputs
2. Enter the file names to use the results of the first step
3. Update flow parameters:
   - Change the Scenario to “`Custom`” and the Planar Area to `200.0` (note the `.0`)



Visualization of flow outputs

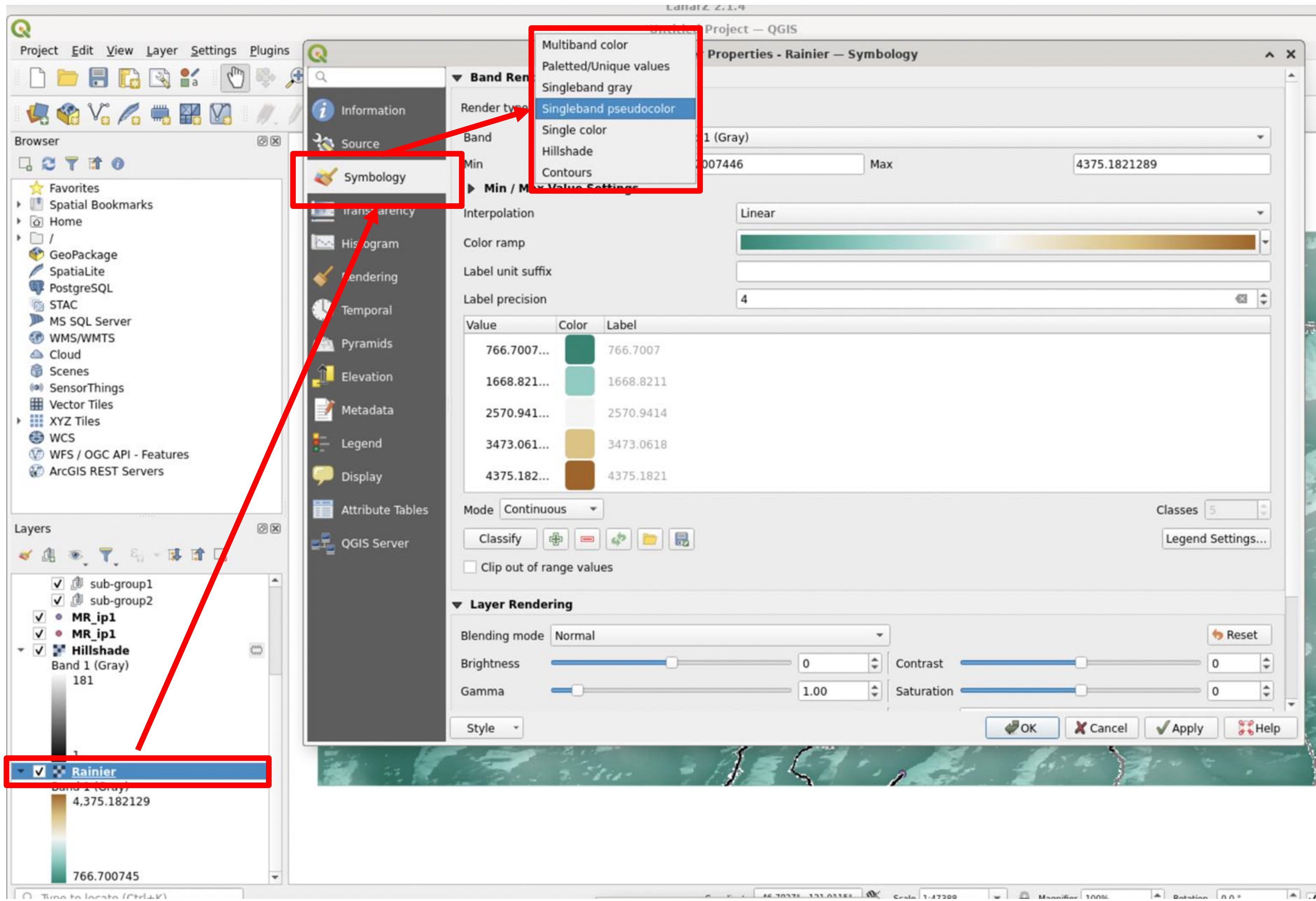
1. Return to your QGIS project
2. Add flow output tiff files from the results folder using `Layer→Add Layer →Add Raster Layer` and choosing the file `total.tif` from the `MR_Lahars` folder



❗ See slides 7-12 for how to change the colors and appearance of the map.

C) Adjust the visualization of the flow outputs: color

1. In the ‘`layers`’ panel at the bottom left of your GQIS screen, double click the `Rainier` layer.
2. A window will appear. From here, ensure you are in the ‘`symbology`’ tab from the list on the left of the window.
3. Change the drop down menu labeled “`Render Type`” at the top of the window to “`Singleband Pseudocolor`”



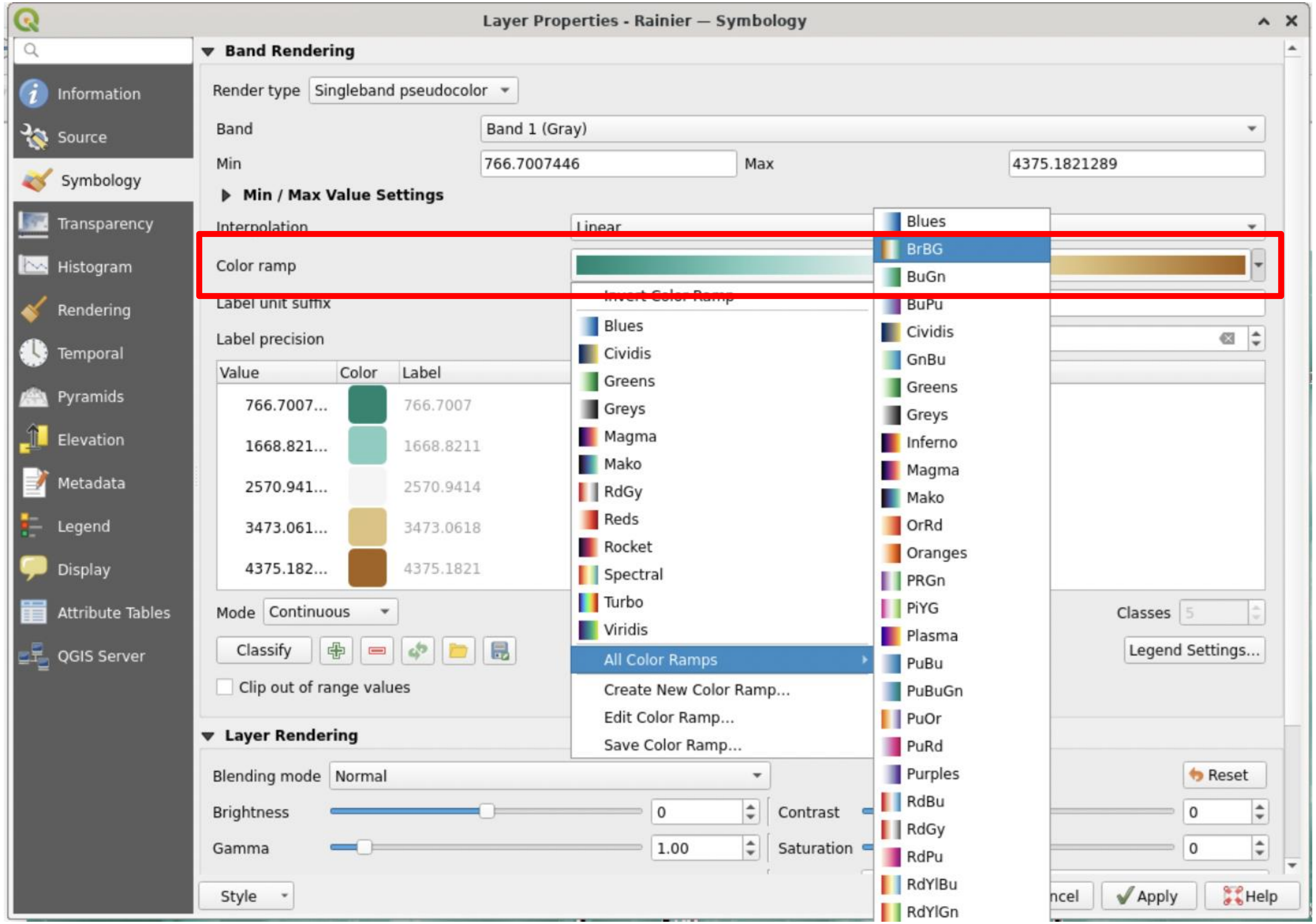
c) Adjust the visualization of the flow outputs: color

4. In order to match the example, click the arrow to the right of the ‘`color ramp`’ option.

5. Hover over ‘`All color ramps`’ from the list that appears and choose “**BrBG.**”

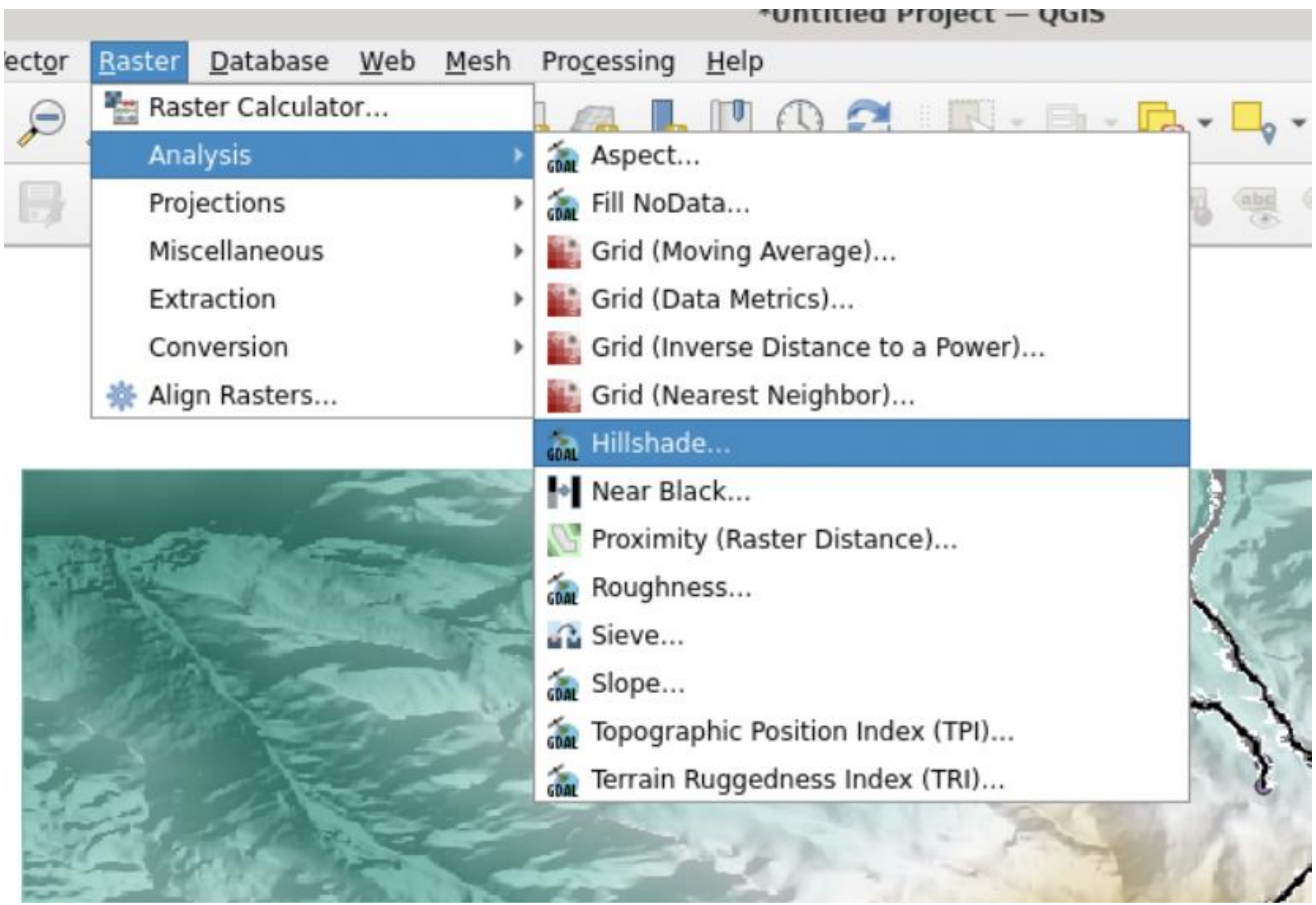
6. Click the arrow to the right of color ramp again and click ‘`Invert Color Ramp`’ at the top of the list that appears.

7. Click apply before closing the window.



d) Adjust the visualization of the flow outputs: topography

1. Navigate to the Raster menu item at the top of the window.
2. Hover over ‘`Analysis`’ and select ‘Hillshade’ from the list that appears.
3. After ensuring the DEM being used is `Rainier.tif`, select ‘run’ at the bottom of the window that appears.

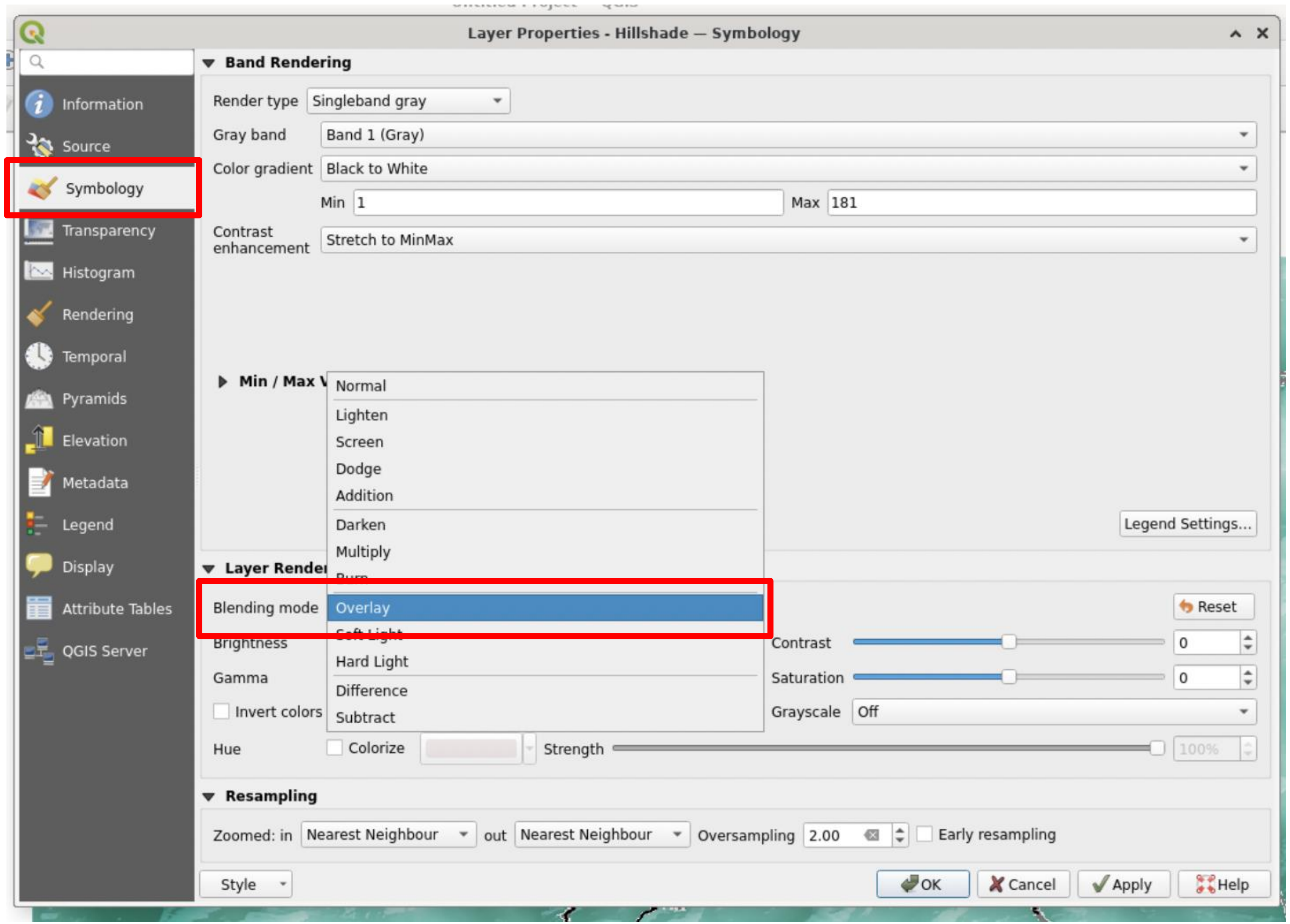


d) Adjust the visualization of the flow outputs: topography

4. Close the window and double click the new ‘`Hillshade`’ layer in the layers panel.

5. Ensure you are in the `symbology` tab. Open the dropdown menu under the “`Blending Mode`” option and select overlay.

6. Click apply at the bottom of the window



d) Adjust the visualization of the flow outputs: topography

7. Open the ‘`Transparency`’ tab in the same window.

8. Adjust the ‘`global opacity`’ slider to around `25%`.

9. Click apply at the bottom of the window.